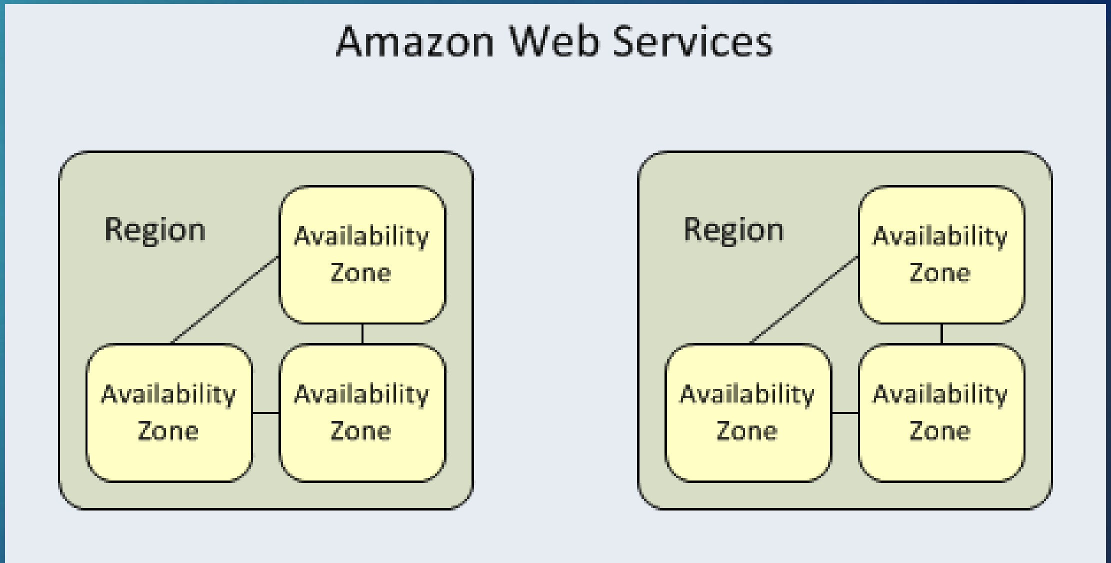
RECAP

- A Region contains 2 more or Availability Zones
- Regions
  - Separate geographic area where your EC2 instances are running
  - Each is designed to be completely isolated from the other to provide fault tolerance and stability
- Availability Zones
  - Isolated, but other zones in same region are connected



RECAP

- Storage models
  - Simple Storage Service (S3)
    - Cheapest for data storage and can be accessed from anywhere via a URL
  - Elastic Block Store (EBS)
    - Mount to a single EC2 instance
    - Only available in a particular region, but faster than S3 and EFS
  - Elastic File System (EFS)
    - Available on multiple regions and can be accessed by up to *thousands* of EC2 instances at once
    - Best for large very large quantities of data such as large analytic workloads
  - Other specialized models:
    - Glacier, Storage Gateway, Snowball, Snowball Edge and Snowmobile

RECAP

- For your S3 website, how much do you think it costs to host per month?
  1. \$1-\$3/month
  2. \$5-\$10/month
  3. \$10-\$20/month
  4. > \$20/month
- Answer
  - Typically, it will cost \$1-3/month

Homework Assignment and Team Project

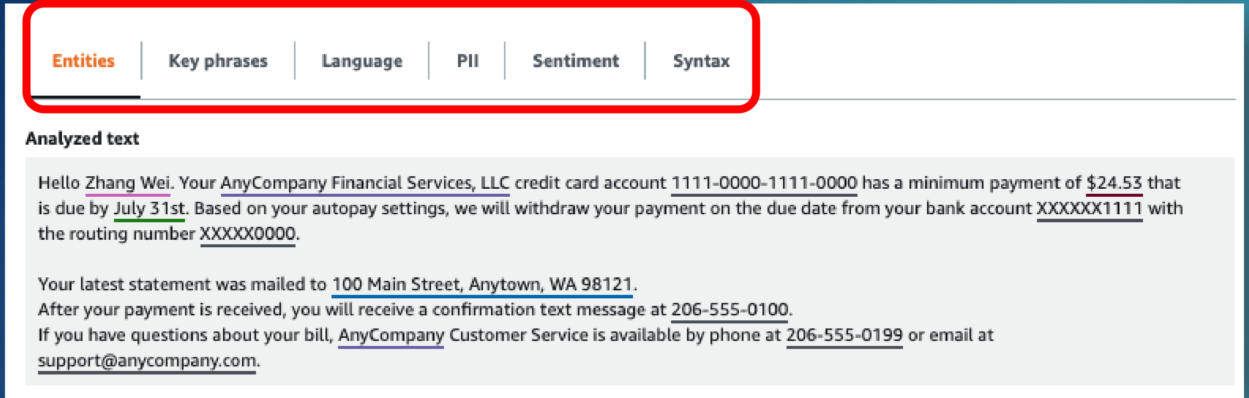
- You will research AWS tutorials of your choice and these will later be shared with the class
- The purpose of the tutorials is to:
  - Get you and your team familiar with AWS services faster to help spark ideas for the team project
  - Share your research with the class to help them better understand technologies they may not have looked at
- We will explore several AWS services in future classes but let's look at a few that won't be covered in class that you may want to research further for your project
  - Several teams used these services on past projects

AWS Comprehend

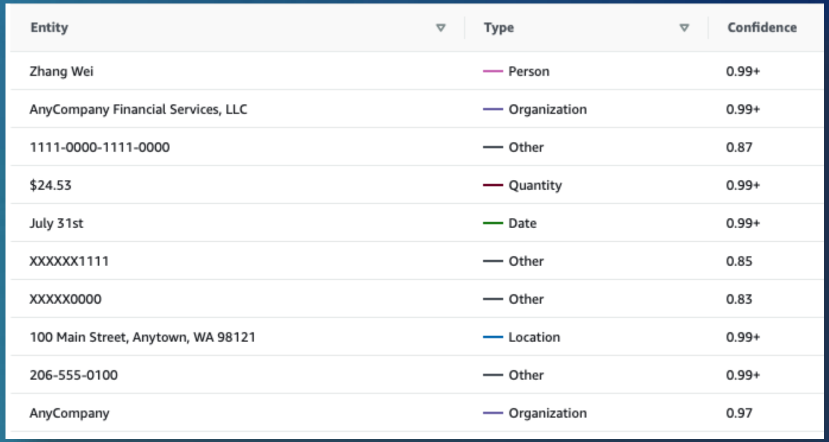
Amazon Comprehend

- A natural-language processing (NLP) service that uses machine learning to uncover information in unstructured data

Input



Output

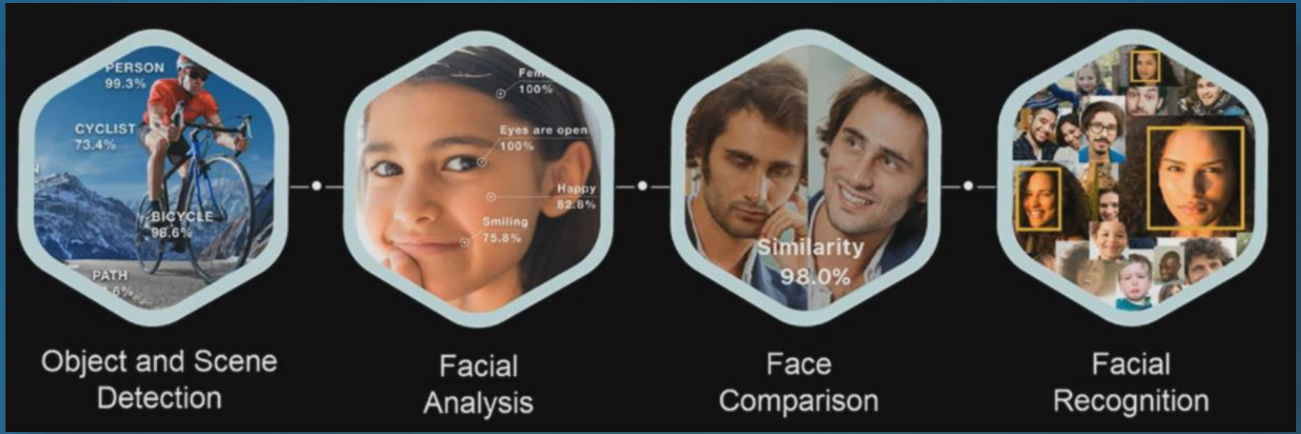


Entity	Type	Confidence
Zhang Wei	Person	0.99+
AnyCompany Financial Services, LLC	Organization	0.99+
1111-0000-1111-0000	Other	0.87
$24.53	Quantity	0.99+
July 31st	Date	0.99+
XXXXXX1111	Other	0.85
XXXXX0000	Other	0.83
100 Main Street, Anytown, WA 98121	Location	0.99+
206-555-0100	Other	0.99+
AnyCompany	Organization	0.97

- Project Ideas:
  - Process a data feed to find out what people are saying about a certain entity (person, topic, etc.)
  - Analyze other content feeds (e.g. News) to look for relationships between common entities

AWS Rekognition

- Identify objects, people, text, scenes, and activities in images and videos, as well as detect any inappropriate content



- Project Ideas
  - Process a live or recorded image feed to determine certain characteristics:
    - How many people are in the image?
    - What other things are in the image (e.g. animal)?

AWS Transcribe

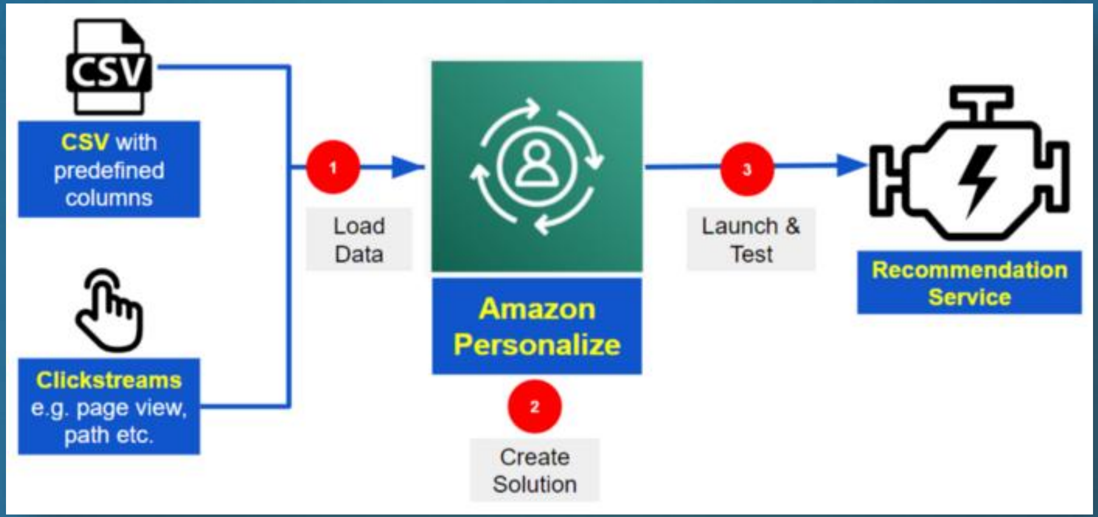
- Uses a deep learning process called automatic speech recognition (ASR) to convert speech to text quickly and accurately
- It is designed to process audio input from a variety of sources such as microphones, audio or video files and provide high quality transcriptions for search and analysis

- Project Ideas:
  - Process recorded audio feed (e.g. lecture) and use the output to create a summarization
  - Process a live audio feed and search for key entities using AWS Comprehend

AWS Personalize

Amazon Personalize

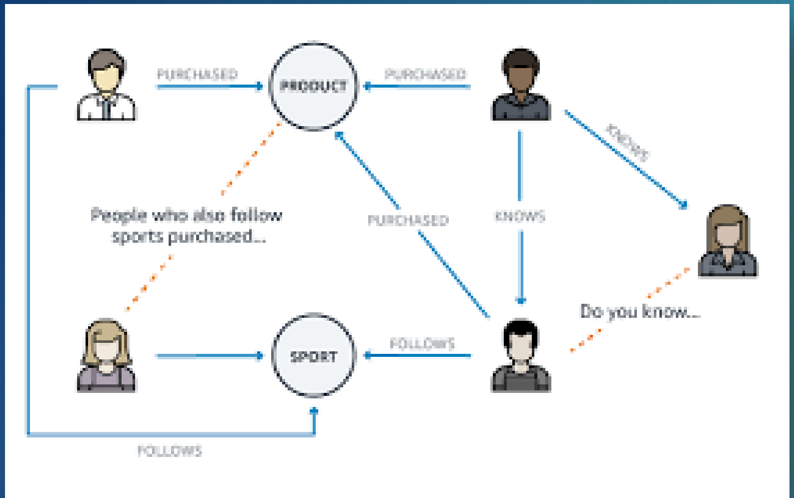
- Enables developers to build applications with the same machine learning (ML) technology used by Amazon.com for real-time personalized recommendations – no ML expertise required

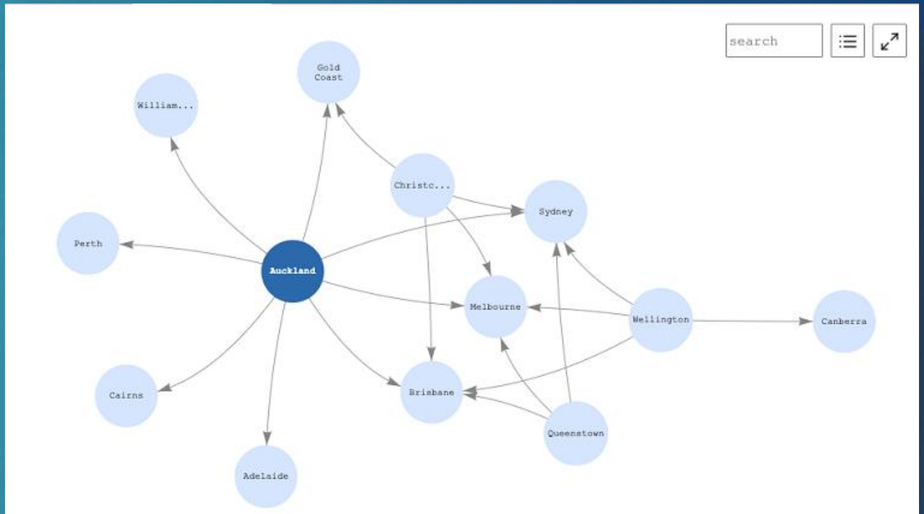


- Project Ideas:
  - Create your own specialized recommendations web site (eStore, travel, restaurant, etc.)
  - Process a resume and recommend relevant jobs using a publicly available Jobs API

AWS Neptune

- Fully managed graph database service that makes it easy to build and run applications that work with highly connected datasets





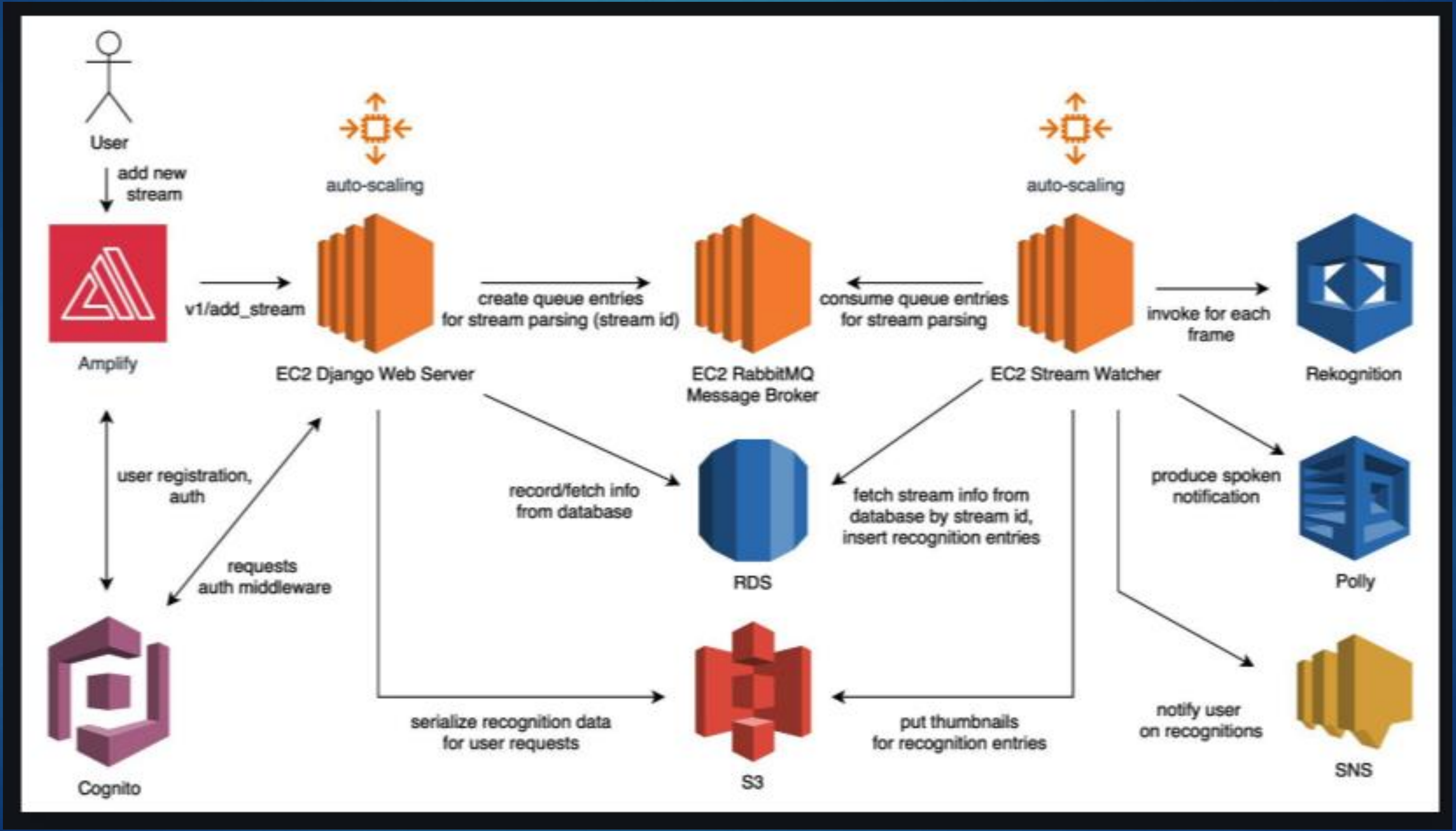
- Project Ideas:
  - Process a data feed and extract common entities (via AWS Comprehend) to visually display a graph
  - Using a data feed of locations of airports, determine the possible routes from one city to another

Other Technologies to Explore

- **Amazon Textract**
  - It goes beyond simple optical character recognition (OCR) to identify, understand, and extract specific data from documents
- **Amazon Polly**
  - Service that turns text into lifelike speech, allowing you to create applications that talk, and build entirely new categories of speech-enabled products
- **Amazon Translate**
  - A neural machine translation service that delivers fast, high-quality, affordable, and customizable language translation
- **Amazon Lex**
  - A service for building conversational interfaces into any application using voice and text

Term Project - Vanessa

Term Project - Vanessa



Cloud Networking Basics

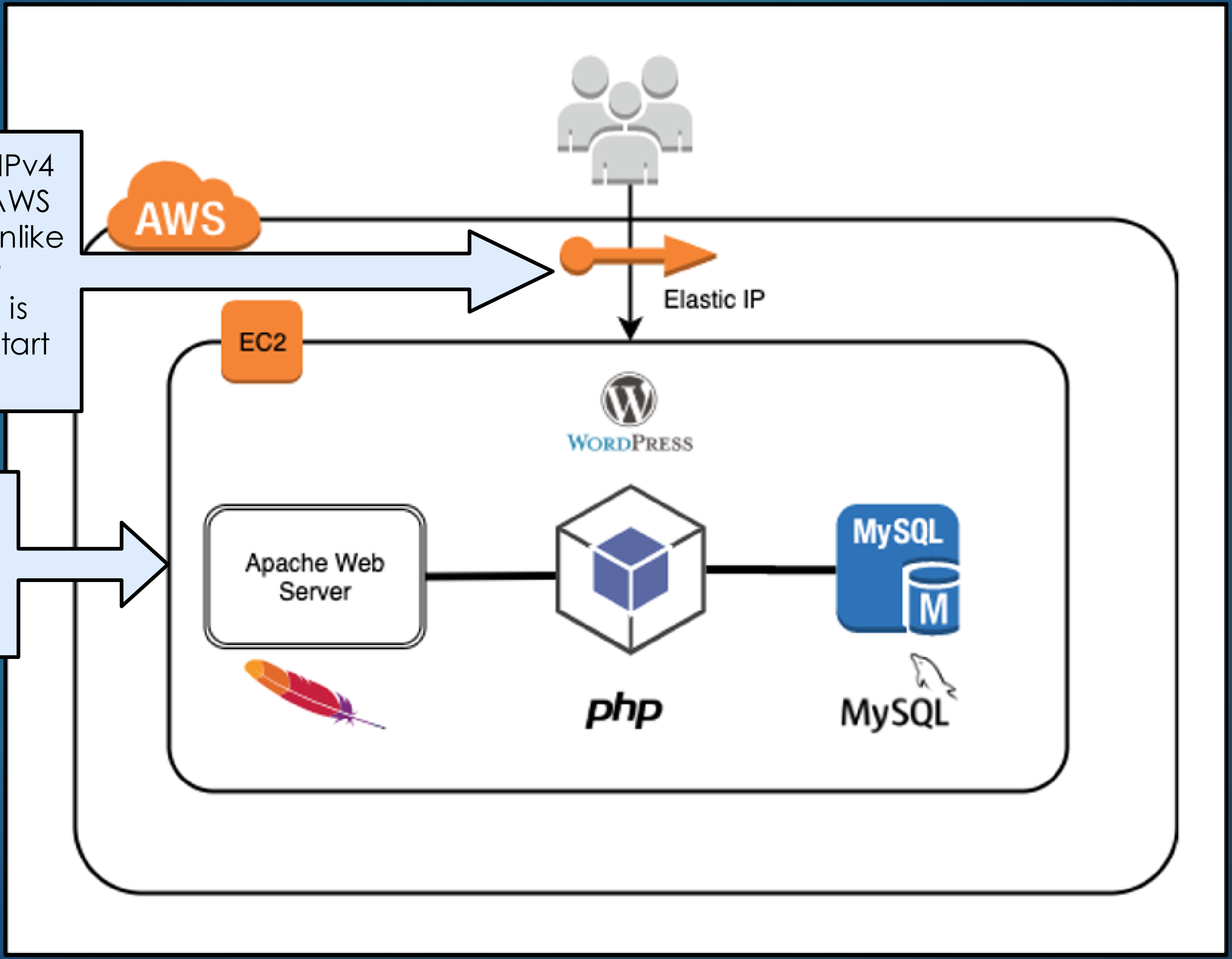
Engineering Cloud Software Systems

Department of Software Engineering
Rochester Institute of Technology

Last Activity – Install WordPress on EC2

An Elastic IP address is a static IPv4 address associated with your AWS account in a specific Region. Unlike the auto-assigned public IP address, an Elastic IP address is preserved after you stop and start your instance

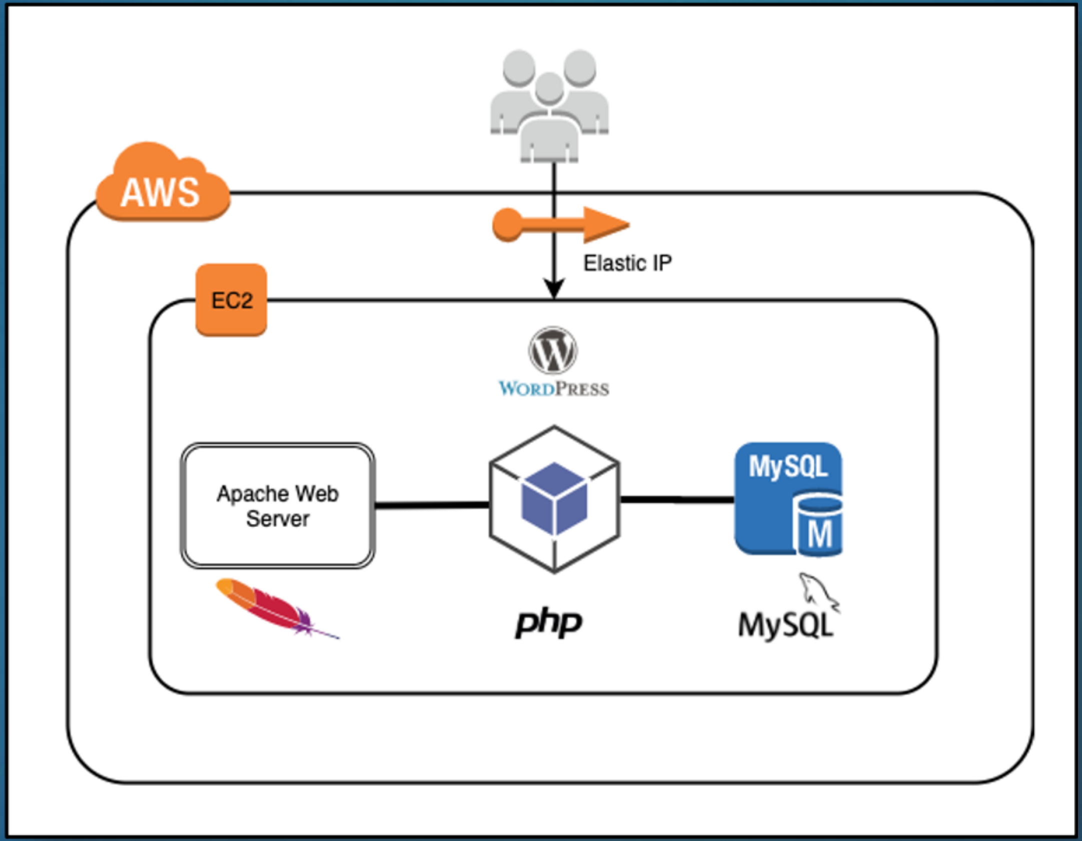
Whenever you start/stop an EC2 instance, AWS will auto-assign a public IP address



Questions:

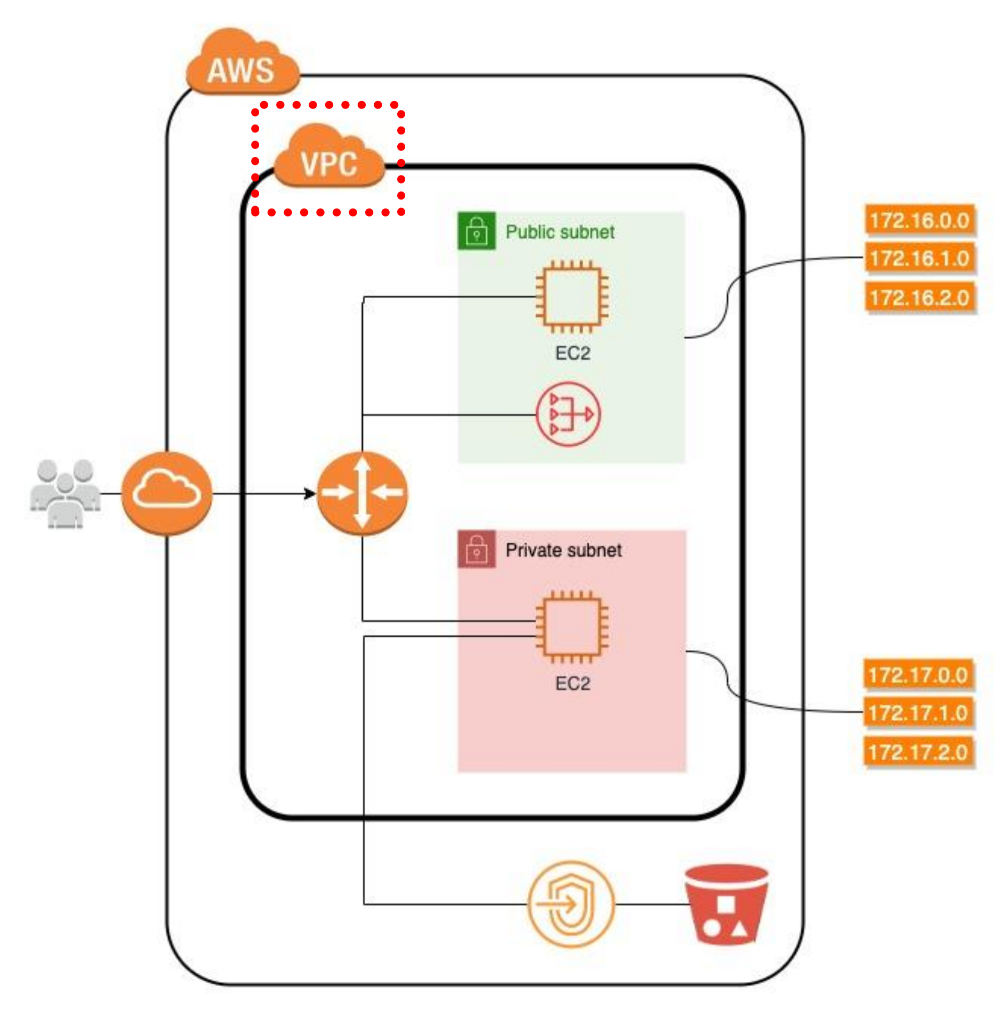
- What if I wanted to upgrade to a larger EC2 instance?
- How can I share my EC2 instance with others?
- How could something like this be developed across a team?
- How does AWS secure my EC2 instance?
- What if I want some of my AWS services (e.g. database) to not be accessible from the Internet?

How does AWS secure my EC2 instance?



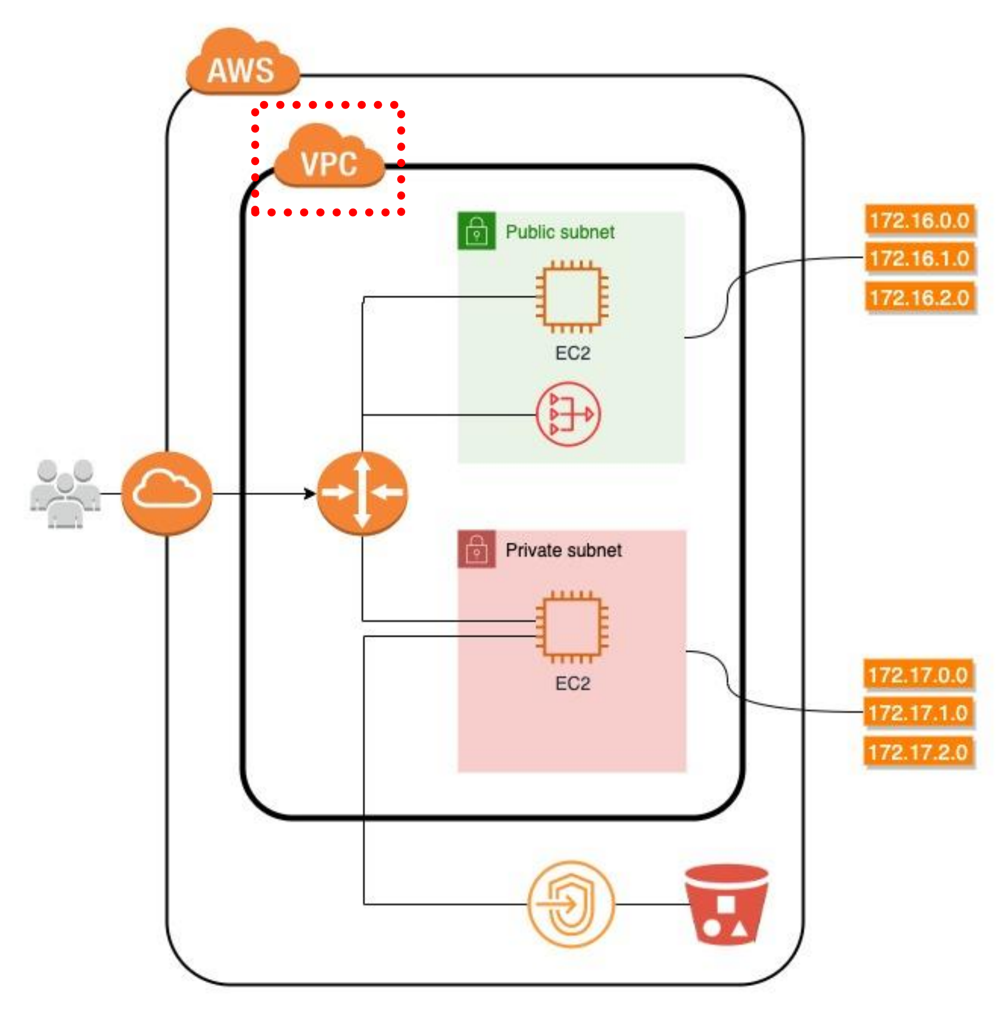
Virtual Private Cloud (VPC) - Overview

- A Virtual Private Cloud (VPC) is a virtual network dedicated to your AWS account
- VPCs are logically isolated from other virtual networks in AWS
- Each VPC exists within a single AWS Region
- A VPC serves as the foundational networking layer for AWS workloads
- Most AWS services operate within the context of a VPC



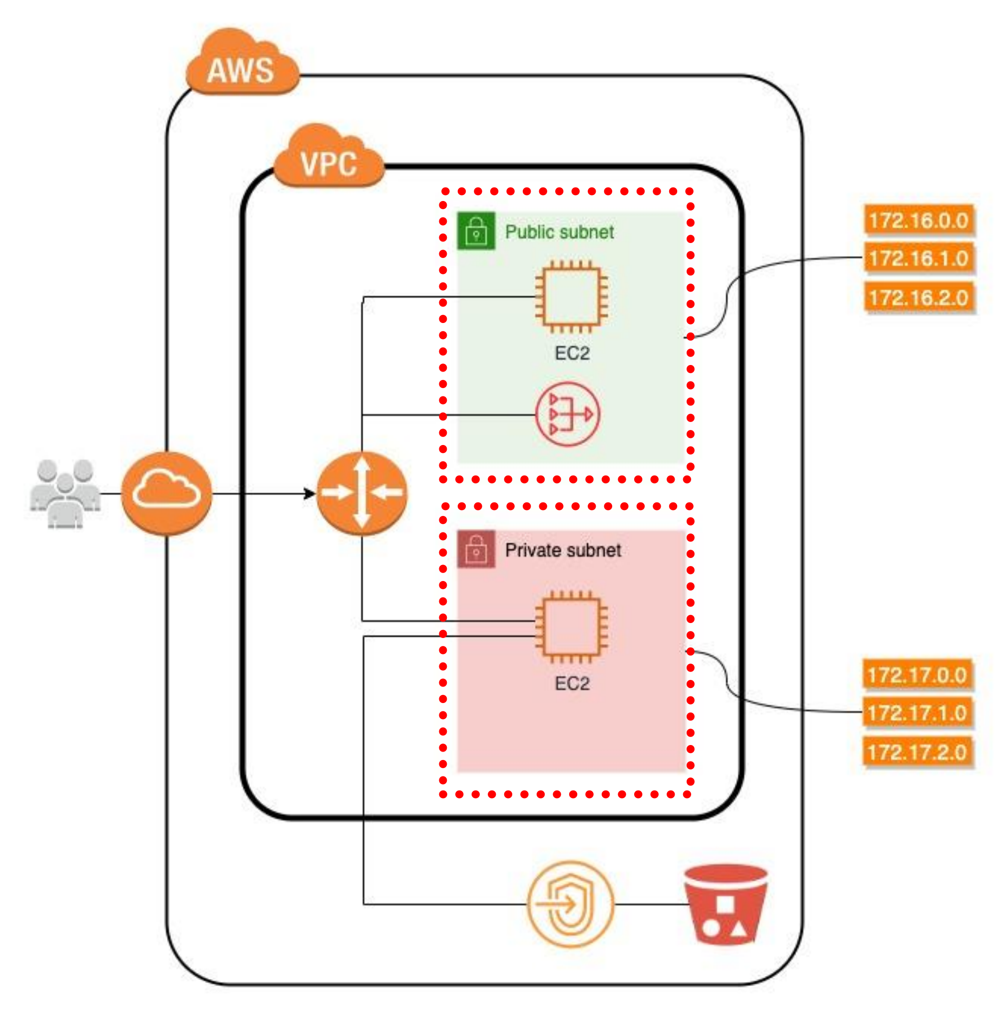
Virtual Private Cloud (VPC) - Purpose

- A VPC defines the **network boundary** for your application in AWS
- It establishes where and how AWS resources can communicate
- VPCs allow isolation of workloads and environments within an account
- Multiple VPCs can exist in a single AWS account for different use cases
- All subsequent networking constructs build upon the VPC foundation



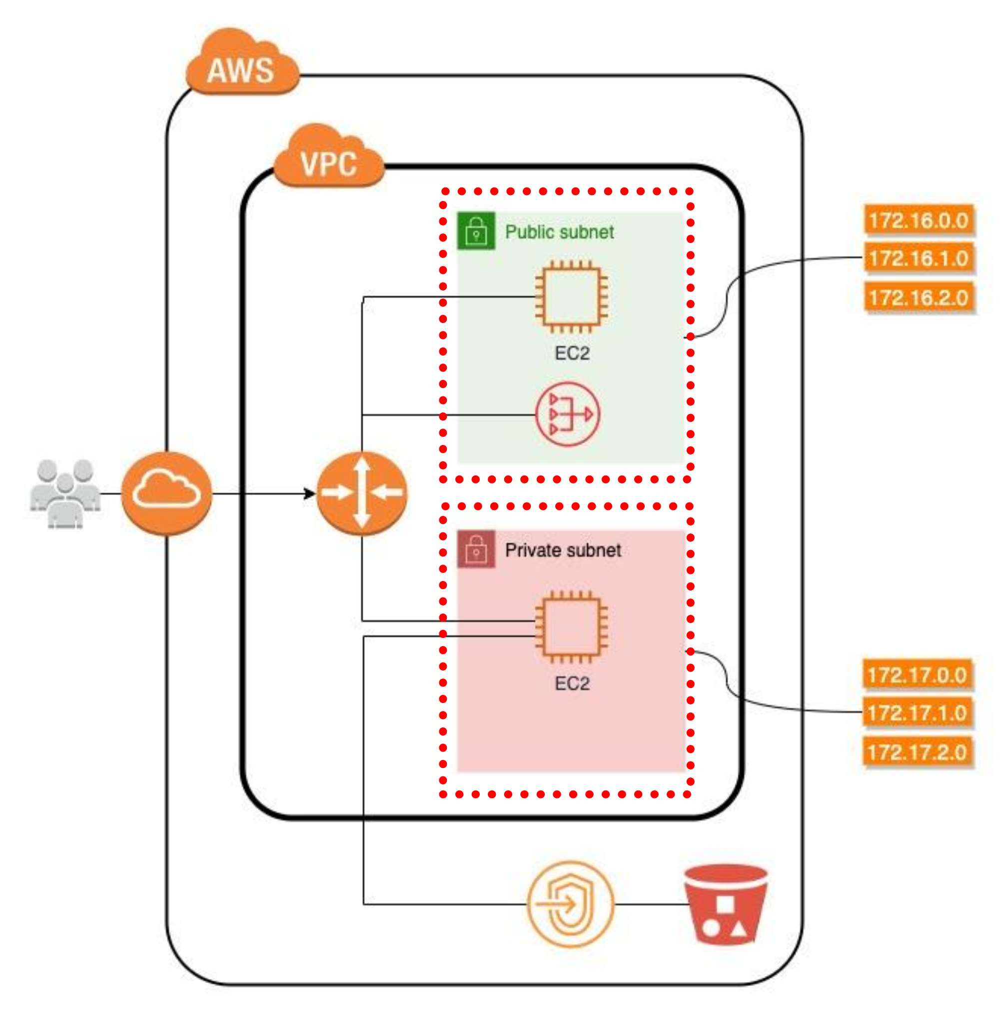
Subnet - Overview

- A subnet is a logical subdivision of a VPC used to organize resources
- Subnets define where AWS resources are placed within a VPC
- Each subnet represents a portion of the VPC's IP address space
- AWS resources such as EC2 instances and load balancers are launched into subnets
- Subnets provide the structural foundation for separating application components



Subnet – Public vs. Private

- A public subnet is intended for resources that must be reachable from outside the VPC
  - They commonly host entry-point components such as load balancers
- A private subnet is intended for resources that should not be directly accessible from the internet
  - They commonly host internal components such as application servers or databases
- Whether a subnet is public or private is determined by its network configuration, not by the resources themselves



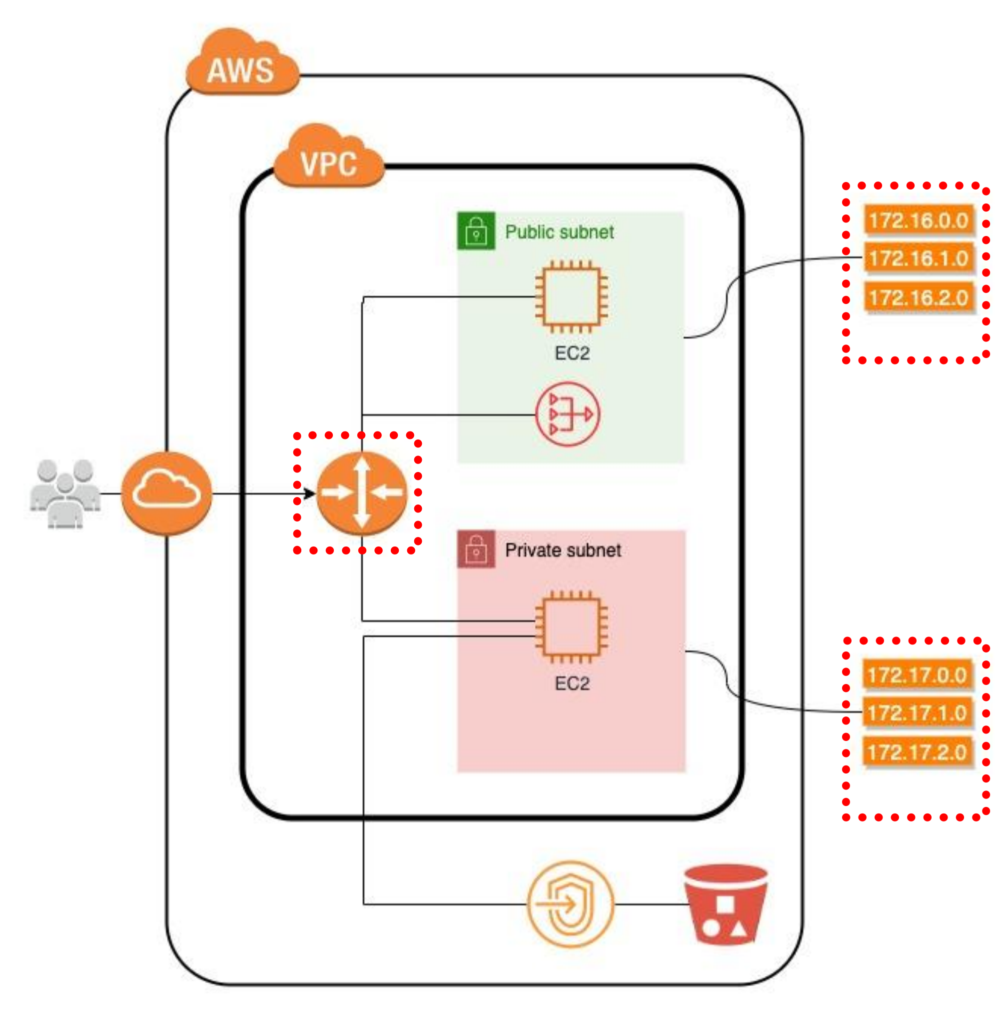
Calculating IP addresses for VPC vs. Subnets

- When you create a VPC, you must specify an IPv4 CIDR (Classless Inter-Domain Routing) block for the VPC. Examples:
  - 10.0.0.0/24 ➔ $2^{32} – 2^{24} = 2^{8}$ = **256** IP addresses for your VPC (10.0.0.0 – 10.0.0.255)
  - 10.0.0.0/20 ➔ $2^{32} – 2^{20} = 2^{12}$ = **4096** IP addresses for your VPC (10.0.0.0 – 10.0.15.255)
  - The allowed block size is between a /16 netmask (65,536 IP addresses) and /28 netmask (16 IP addresses)
- After you define your VPC, you define your subnets, which are within the range of the VPC. For example, a VPC with 2 subnets:

Network Component	CIDR	# of IP addresses	IP Range
VPC	10.0.0.0/24	256	**10.0.0.0 – 10.0.0.255**
Subnet #1	10.0.0.0/26	64	10.0.0.0 - 10.0.0.63
Subnet #2	10.0.0.64/26	64	10.0.0.64 - 10.0.0.127

Route Table - Overview

- A route table defines how network traffic is directed within a VPC
- It contains a set of rules, called routes, that control traffic flow
- Route tables determine where traffic from a subnet is sent
- Every subnet in a VPC must be associated with a route table
- Route tables are a core component of how AWS networking operates

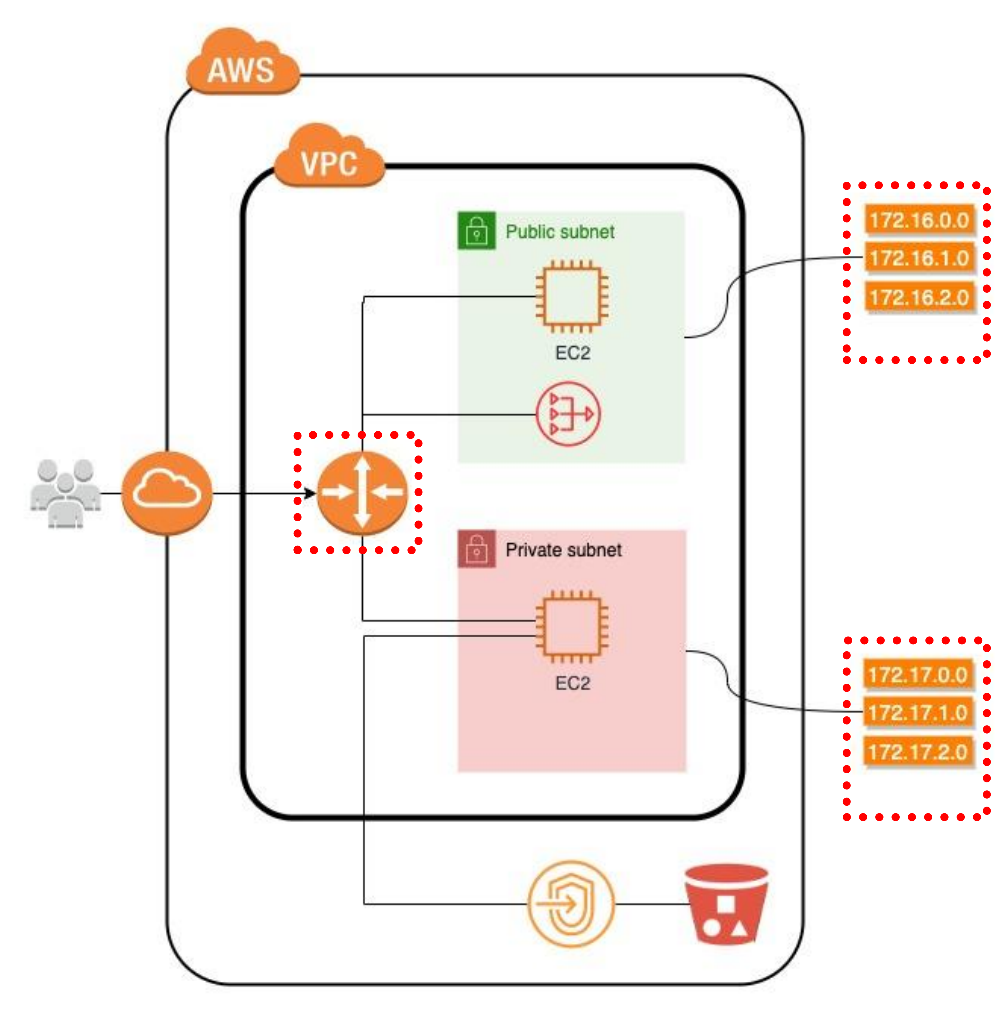


Route Table – Rules and Internet Access

- Each route consists of a destination and a target
- The destination specifies the IP address range for the traffic
- The target specifies where that traffic should be sent
- To allow outbound internet access, a common route is:
  - **Destination:** 0.0.0.0/0 (all IPv4 addresses)
  - **Target:** Internet Gateway attached to the VPC

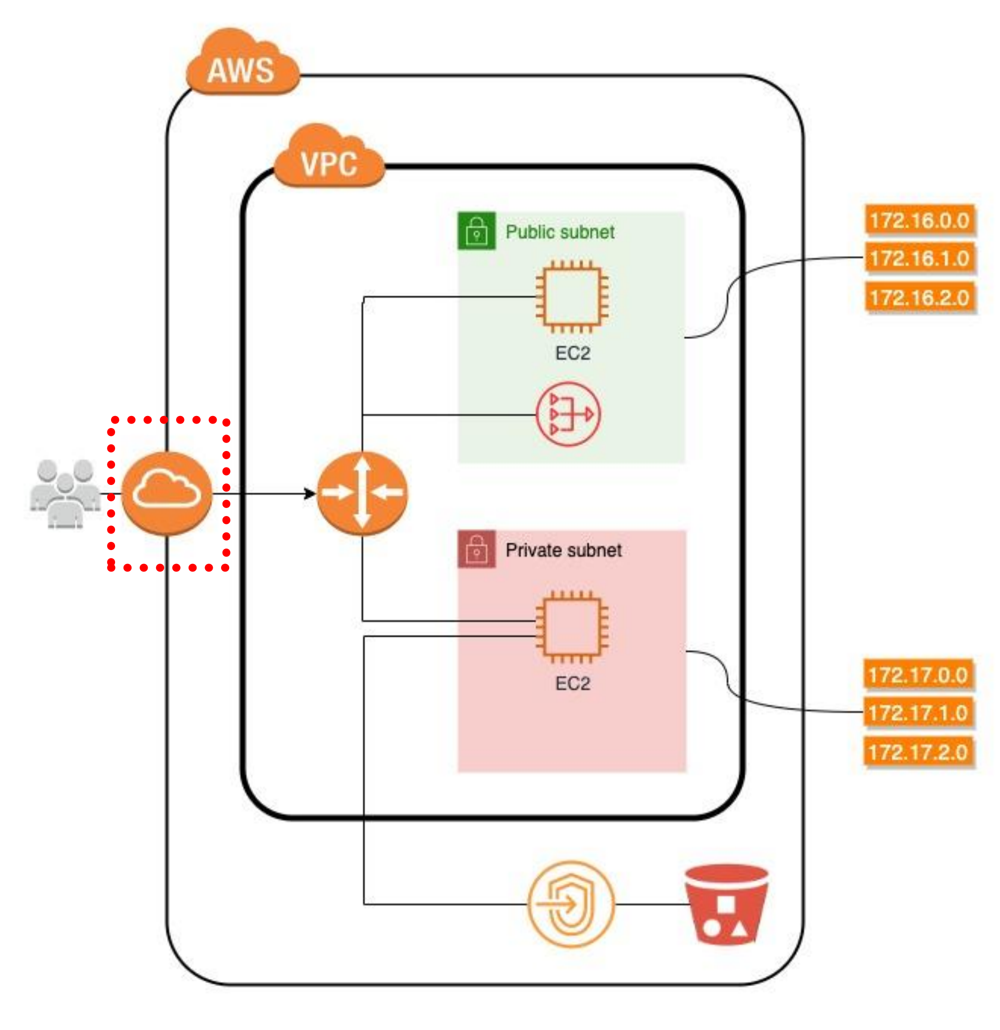
Destination	Target
0.0.0.0/0	igw-12345678901234567

- Route tables are what ultimately determine whether a subnet is public or private



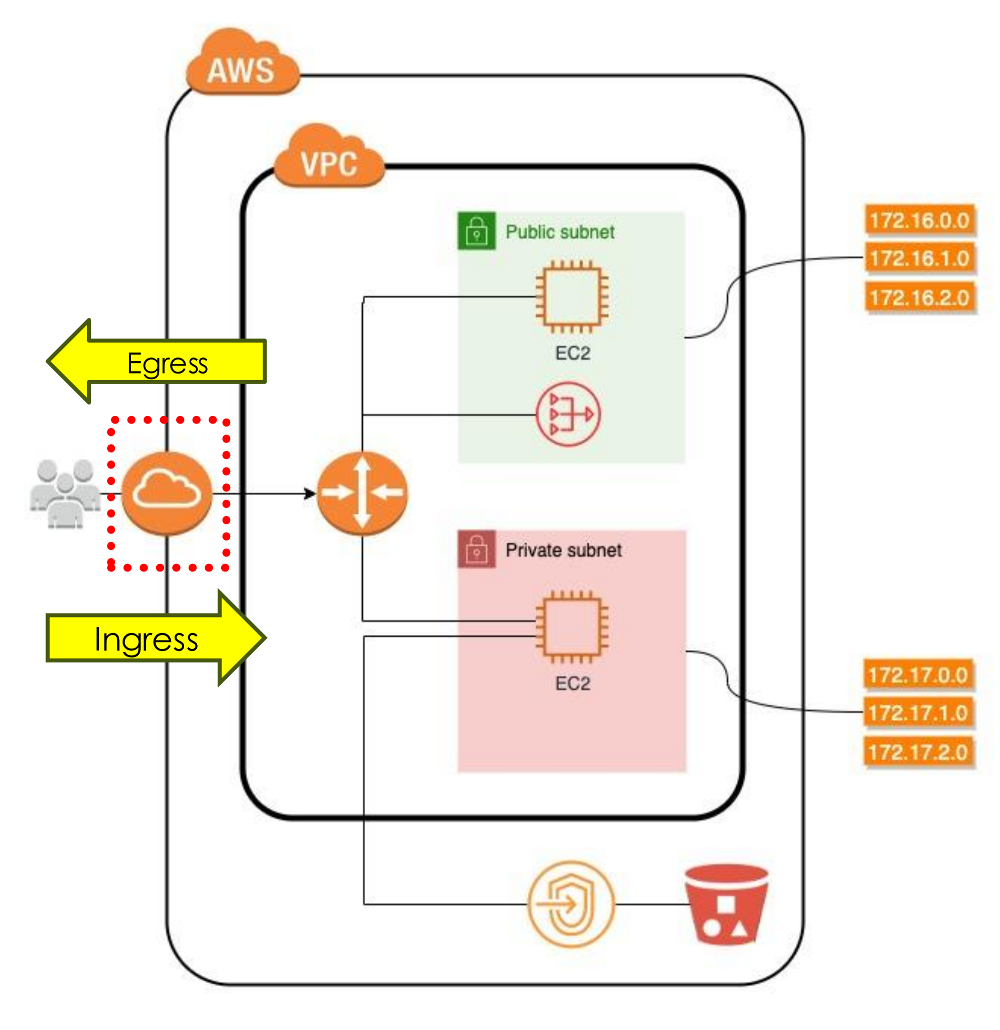
Internet Gateway - Overview

- An Internet Gateway (IGW) enables communication between a VPC and the public internet
- It provides a connection point that allows resources in a VPC to send and receive internet traffic
- An IGW is attached at the VPC level, not to individual subnets or resources
- Each VPC can have only one Internet Gateway attached at a time
- An Internet Gateway must be used in conjunction with route tables to allow internet access



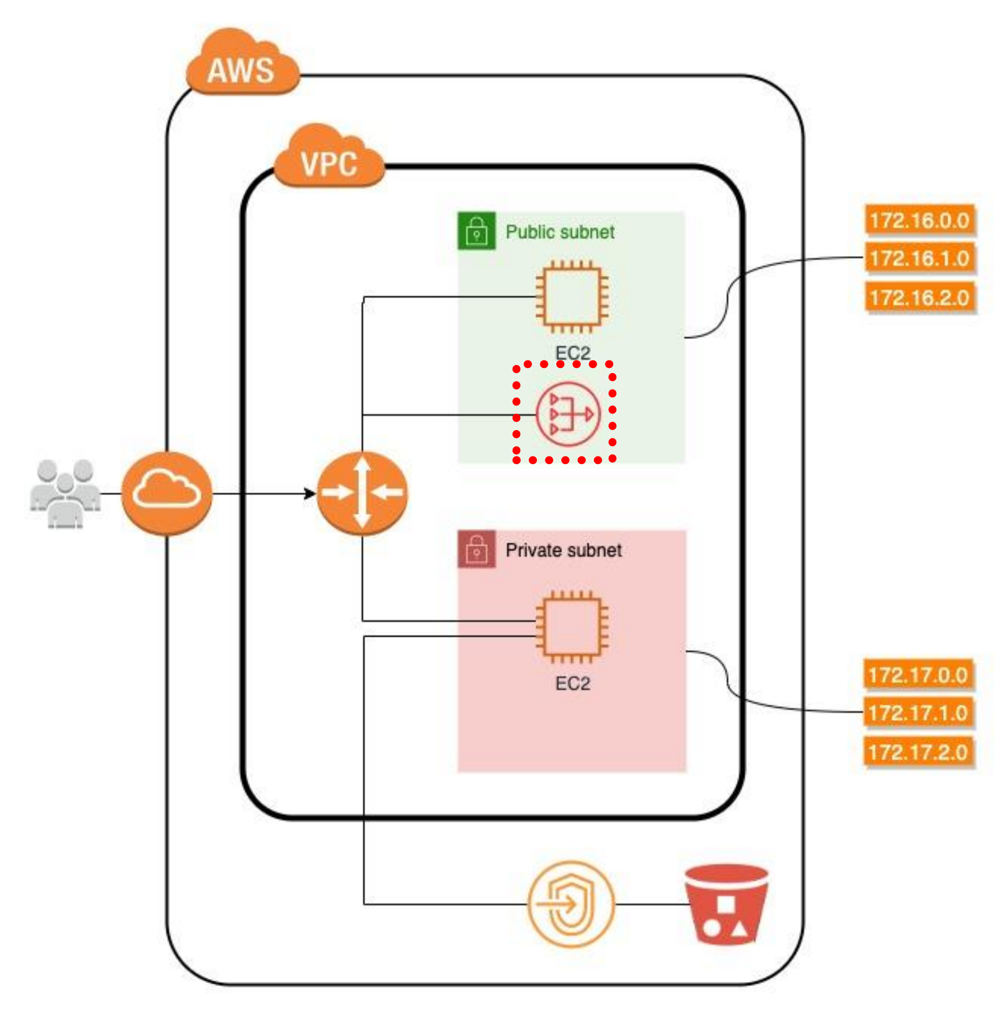
Internet Gateway – Ingress and Egress Traffic

- Ingress refers to network traffic entering the VPC from the internet
- Egress refers to network traffic leaving the VPC to the internet
- An Internet Gateway supports both ingress and egress traffic
- Whether ingress or egress traffic is allowed depends on route tables and security controls
- Internet access is explicitly configured and not enabled by default



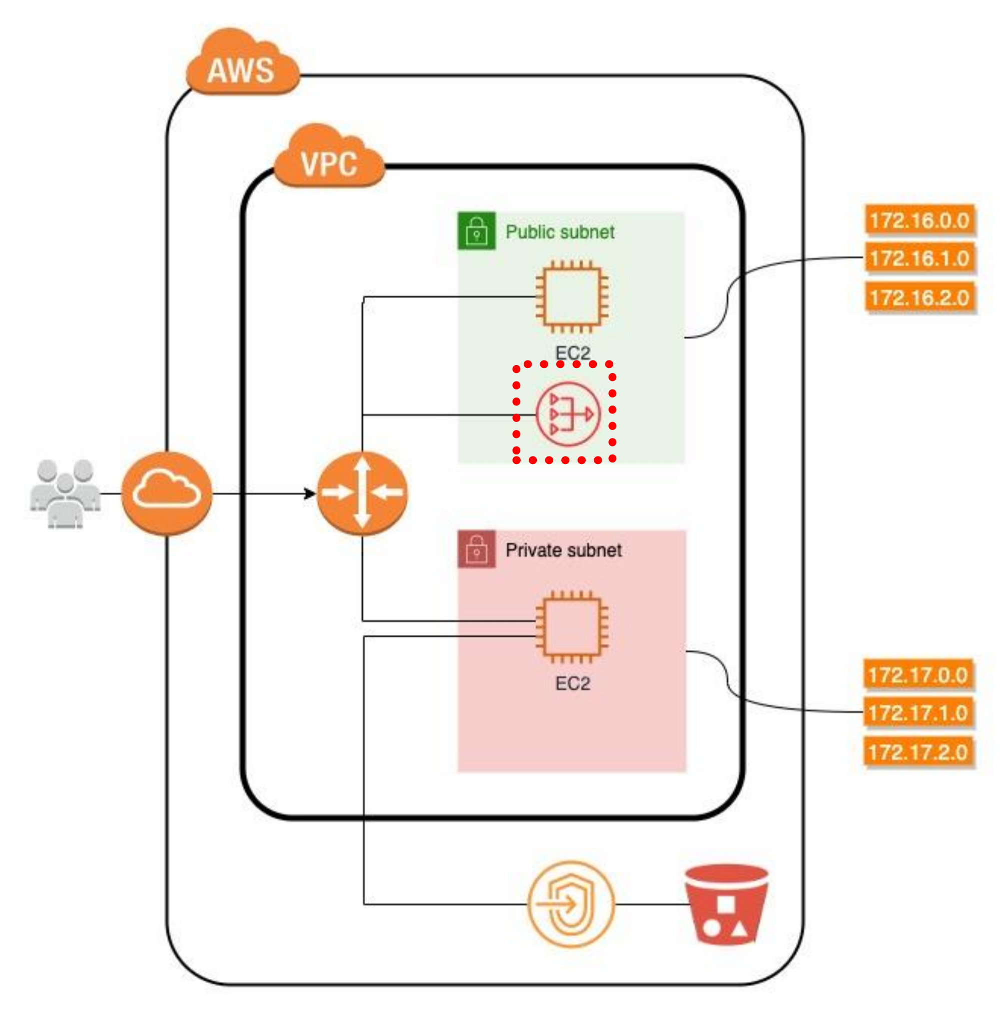
Network Address Translation (NAT) Gateway

- A NAT Gateway allows resources in a private subnet to initiate outbound connections to the internet or AWS services
- It enables egress-only internet access for private resources
- Internet traffic cannot initiate inbound connections to resources behind a NAT Gateway
- NAT Gateways are commonly used for private application servers that need external access
- NAT Gateways improve security by keeping private resources unreachable from the internet



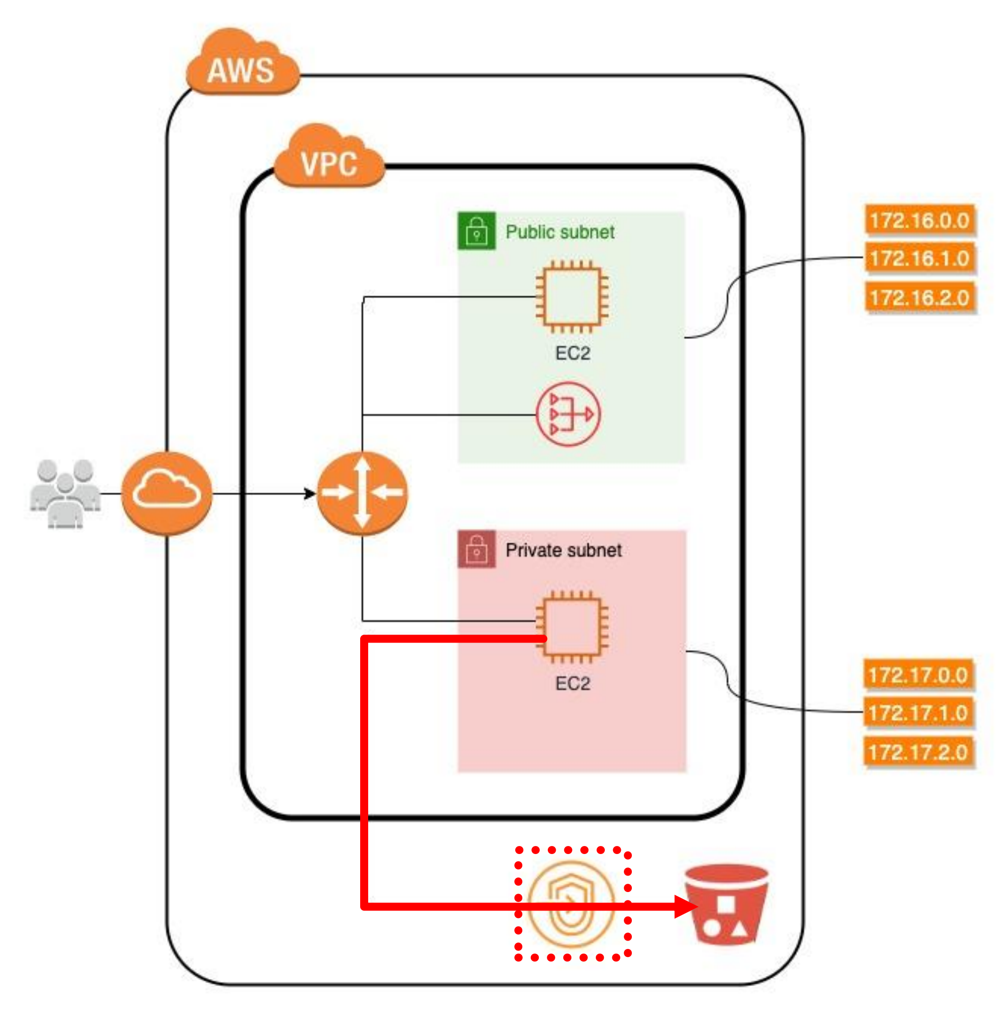
Network Address Translation (NAT) Gateway

- A NAT Gateway must be deployed in a **public subnet**
- The public subnet must have a route to an Internet Gateway
- An **Elastic IP address** is required and associated with the NAT Gateway
- Private subnets route outbound internet traffic to the NAT Gateway via route tables
- NAT Gateways are managed, highly available AWS services



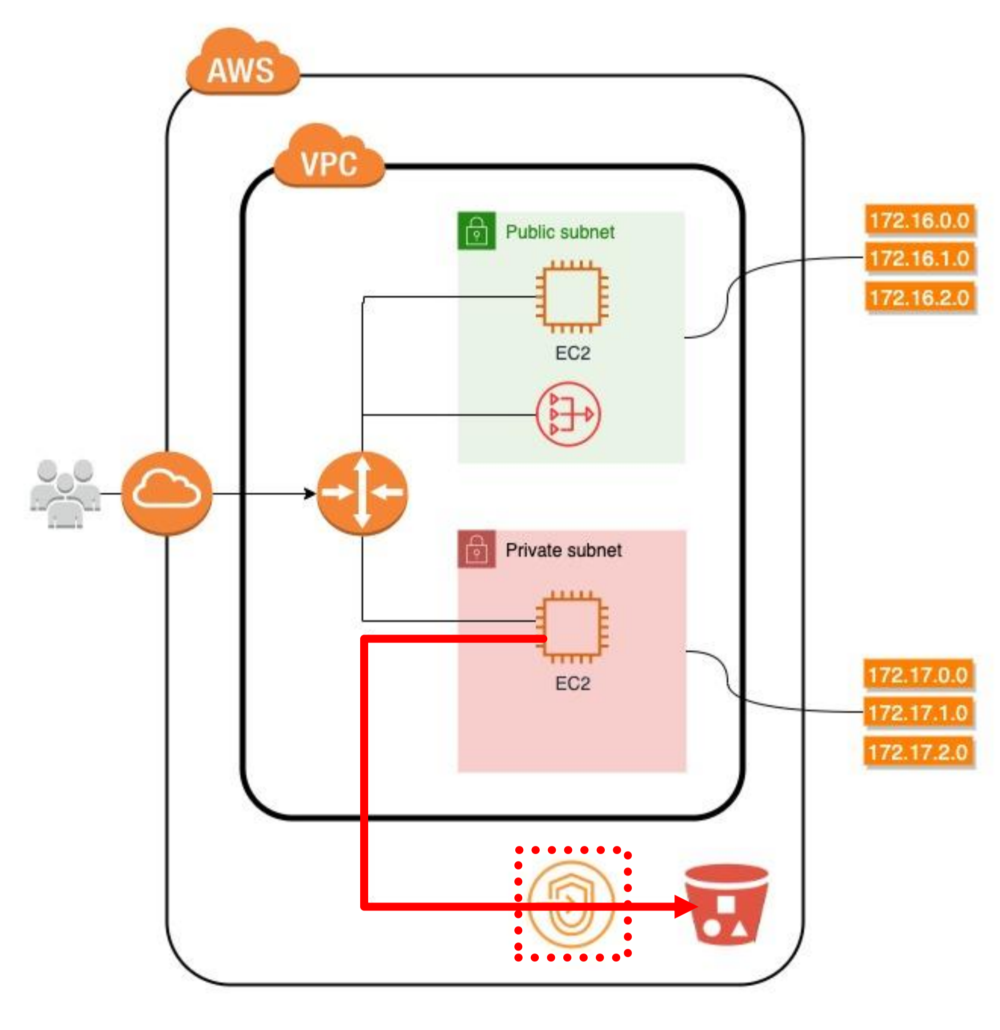
VPC Endpoint - Overview

- A VPC Endpoint enables private connectivity between a VPC and supported AWS services
- Traffic between the VPC and AWS services **does not traverse the public internet**
- VPC Endpoints allow access to services such as Amazon S3 and DynamoDB without using an Internet Gateway or NAT Gateway
- Connectivity occurs entirely within the AWS network
- VPC Endpoints are used to support private, secure architectures

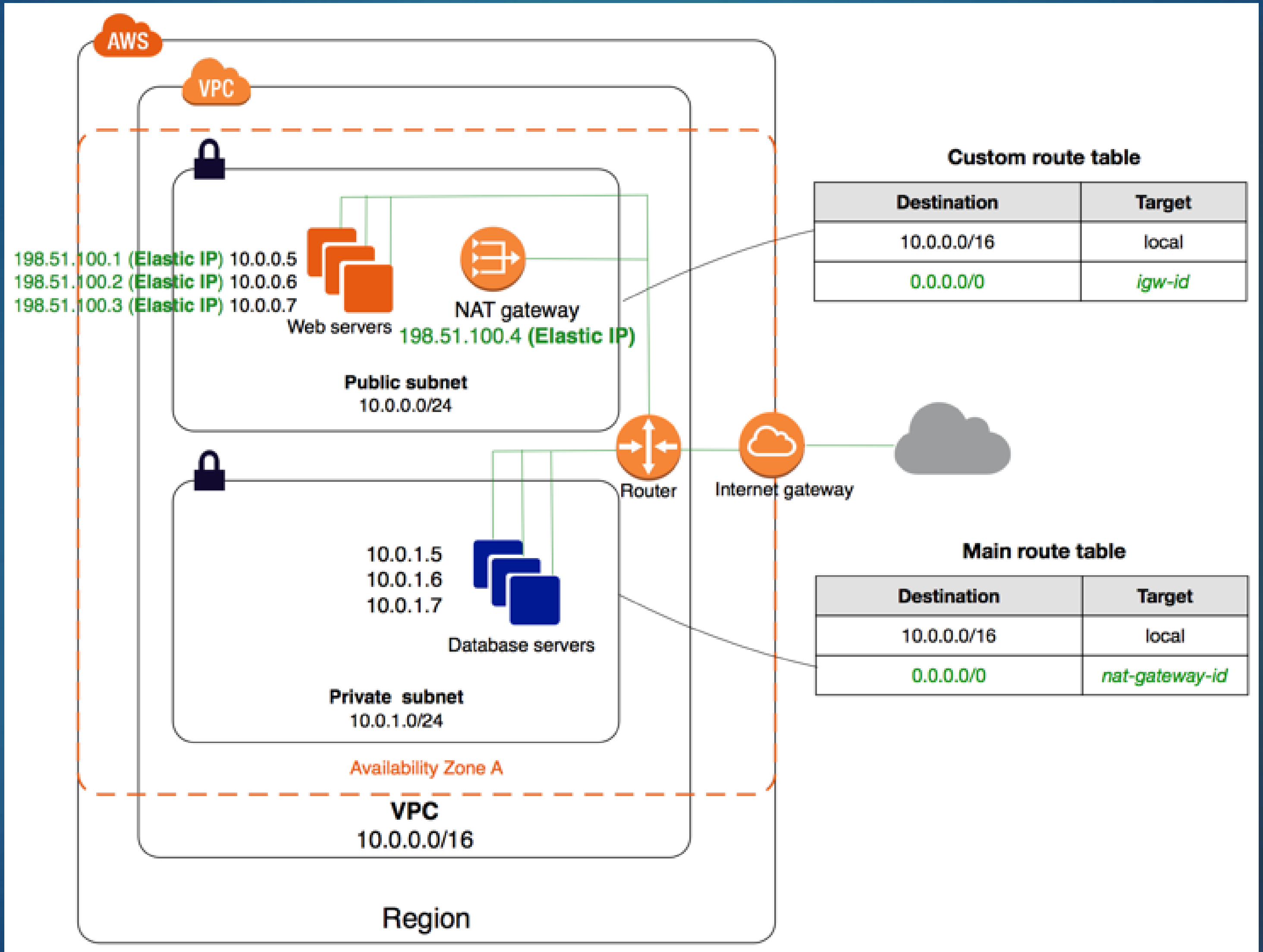


VPC Endpoint – Key Benefits

- Improves **network security** by keeping traffic off the public internet
- Reduces dependency on Internet Gateways and NAT Gateways
- Can lower costs by minimizing NAT Gateway data processing charges
- Commonly used for private workloads accessing AWS-managed services
- Often combined with private subnets to build fully private architectures



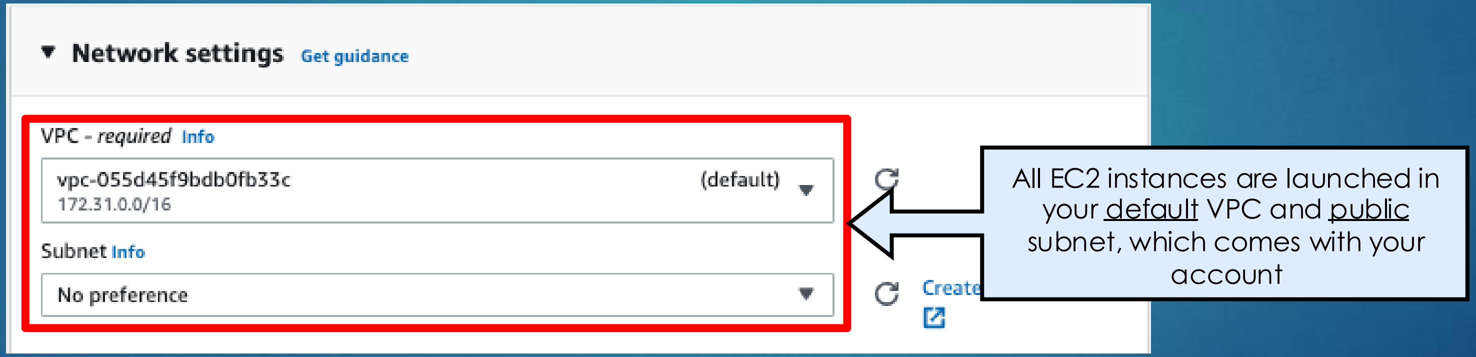
Tying this all together…



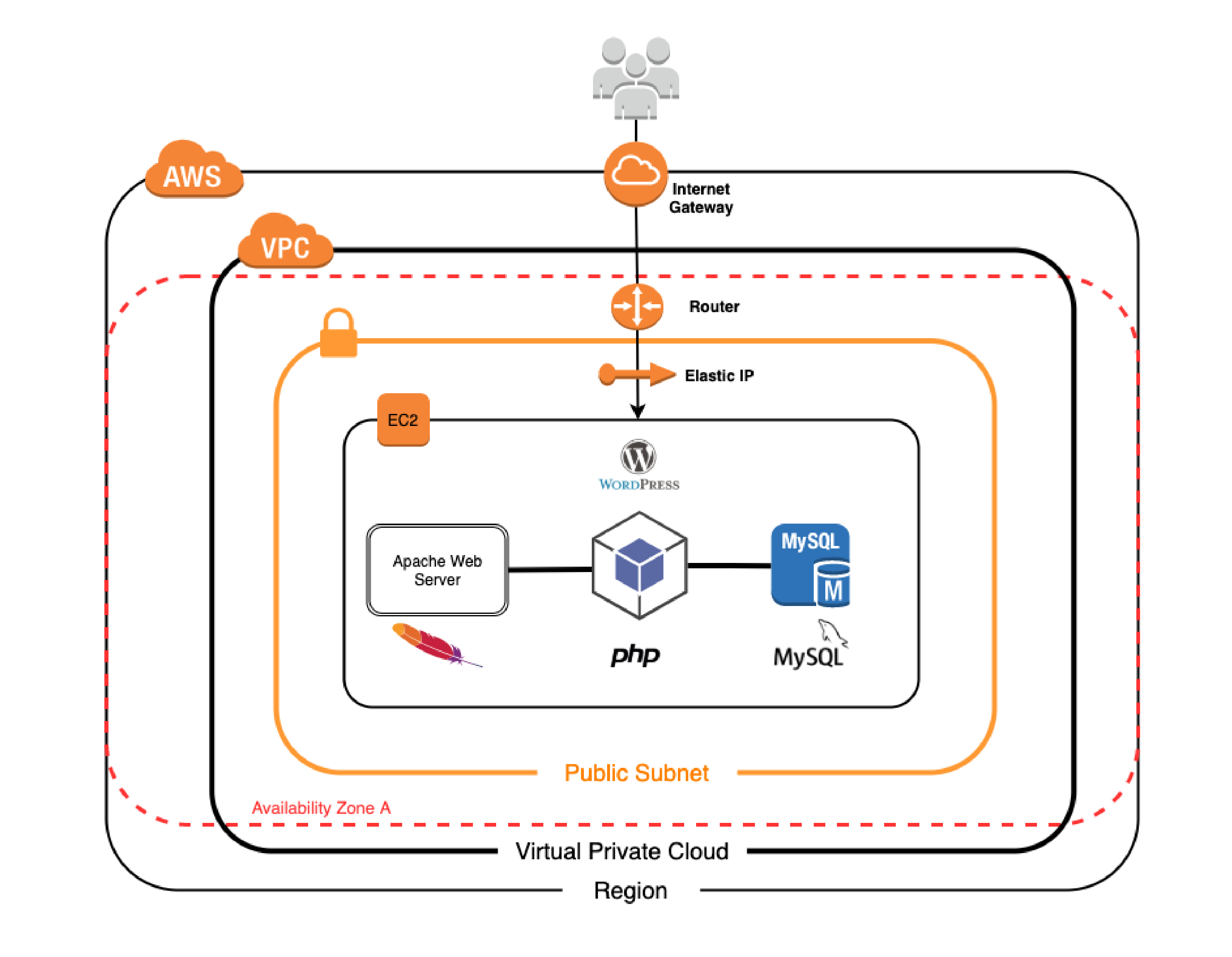
Source: https://gruntwork.io/guides/networking/how-to-deploy-production-grade-vpc-aws/#vpc_peering

VPC and Subnets

- For the WordPress EC2 instance you created, what subnet was it deployed to:
  - Public
  - Private
  - We haven’t created it yet



VPC and Subnets



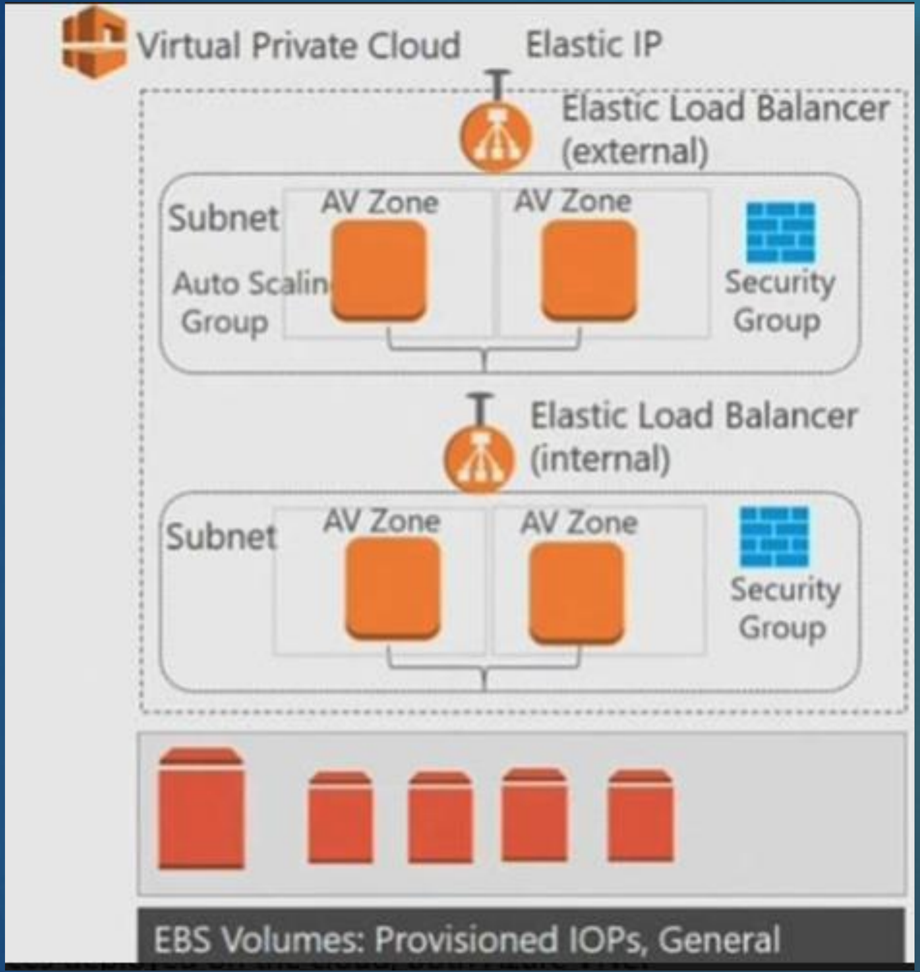
A VPC is located in a Region

Within a Region there are 2 or more Availability Zones

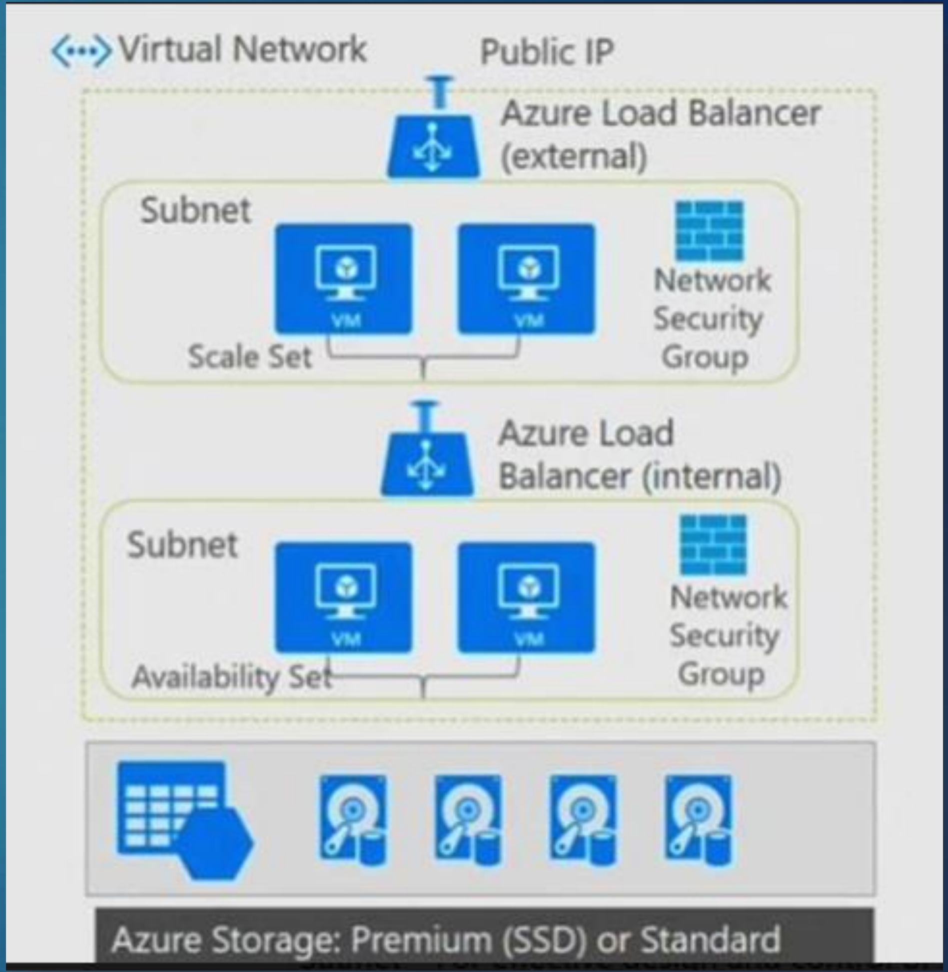
A Subnet is assigned to an Availability Zone

EC2

VPC and Subnets – AWS vs. Azure



AWS



Azure

Source: [Differentiating between Azure Virtual Network (VNet) and AWS Virtual Private Cloud (VPC)]

Class Activity

- Go to myCourses to Assignments > Activity – Creating a VPC and Subnets
  - Remember to take a screenshot when you are done